BTS-VIII(S)-07.19-0899 Reg.No. [| | | | | | |]

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B.Tech. Degree VIII Semester Supplementary Examination July 2019

CS 15-1801 ADVANCED ARCHITECTURE AND PARALLEL PROCESSING

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) What are the memory update options in PRAM?
(b) Explain any 2 static connection networks
(c) Differentiate between RISC and CISC processor architectures
(d) Explain the terms: temporal locality, spatial locality, sequential locality
(e) What is meant by sequential consistency model? Give Lamport's definition for sequential consistency.
(f) Explain asynchronous pipeline model.
(g) What is meant by cache coherence problem? What are the sources of this problem?
(h) Explain wormhole routing
(i) Differentiate between synchronous and asynchronous message passing
(j) Explain ***two-bit predictor*** branch prediction method

PART B

$(4 \times 10 = 40)$

II. Classify the memory models in multiprocessors (10)

OR

III. (a) Consider the following segment (7)

S1: Load R1, A
S2: Add R2, R1
S3: Move R1, R3
S4: Store B, R1

Identify the data dependencies and explain them. Also, draw the dependency graph

(b) Suppose that we are considering an enhancement that runs 10 times faster than original machine, but it is usable only 40% of the time. What is the overall speedup gained by incorporating the enhancement? (3)

IV. Write notes on Superscalar, VLIW and vector processors. (10)

OR

V. Explain backplane bus specification (10)

(P.T.O.)

VI. (a) Consider a 4-segment pipeline with stage delays 2ns, 8ns, 3ns, 10ns. Find the time taken to execute 100 tasks in the above pipeline. (3)

(b) Consider a 4-stage pipeline and the time taken by the four stages are 50ns, 60ns,110ns and 80ns respectively. Pipeline registers are required after every pipeline stage and each of these pipeline registers consumes 10ns delay. What is the speedup of the pipeline under ideal conditions compared to the corresponding non-pipelined implementation? (4)

(c) What is the minimum number of stages needed in a pipelined processor when its speedup is 4.8 and efficiency is 76%? (3)

OR

VII. Describe the 2 types of SIMD computer models (10)

VIII. Explain any 2 parallel programming models (10)

OR

IX. Explain the model of a typical processor that supports instruction level parallelism. (10)

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